



S.O.F.A.R

(SCAM ONLINE FORENSIC AGENT RESPONDER)

TRACK 5: SECURE DIGITAL (FINTECH & SECURITY)



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MY TEAM



NEO LI XIN

Team Leader



CHEW CHIU XIAN

UI/UX Developer



TAN LI JI

ADK Developer

Cybercrime is Accelerating

- **2024:** 35,368 cases | **RM 1.58B** Lost
- **2025 (Q1 Only):** 12,110 cases | **RM 573.7M** Lost
- **The Trend:** On track to exceed **RM 2.2B** this year.

Source: PDRM Official Report





PROBLEMS



THE "GOLDEN HOUR" GAP

Traditional scam reporting in Malaysia often happens too late, allowing mule accounts to be drained before authorities can act.



INFORMATION SILOS

Victims often have to report separately to banks, PDRM, and the NSRC, leading to fragmented evidence.



SOPHISTICATED IMPERSONATION

Scammers use advanced psychological tactics and official-looking phishing links (like fake LHDN or BNM portals) that bypass standard filters.

SOLUTION

Autonomous Multi-Agent Orchestration

Using Genkit to coordinate three specialized agents included Scam Analyst, Forensic Specialist, and Incident Responder to handle a report from start to finish.

Real-Time Forensic Validation

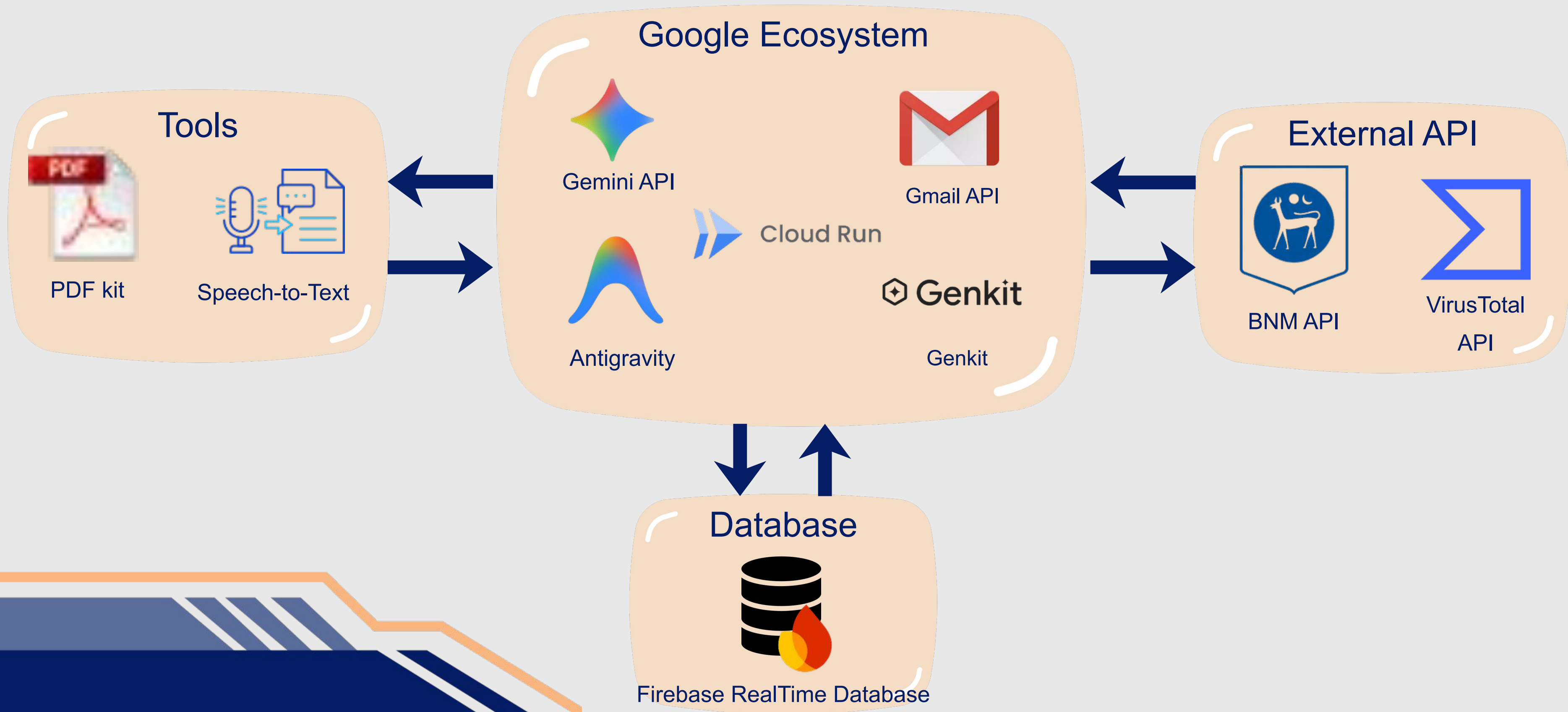
Integrating VirusTotal and SSL checks to provide immediate technical proof of a scam's infrastructure.

Automated Authority Alerting

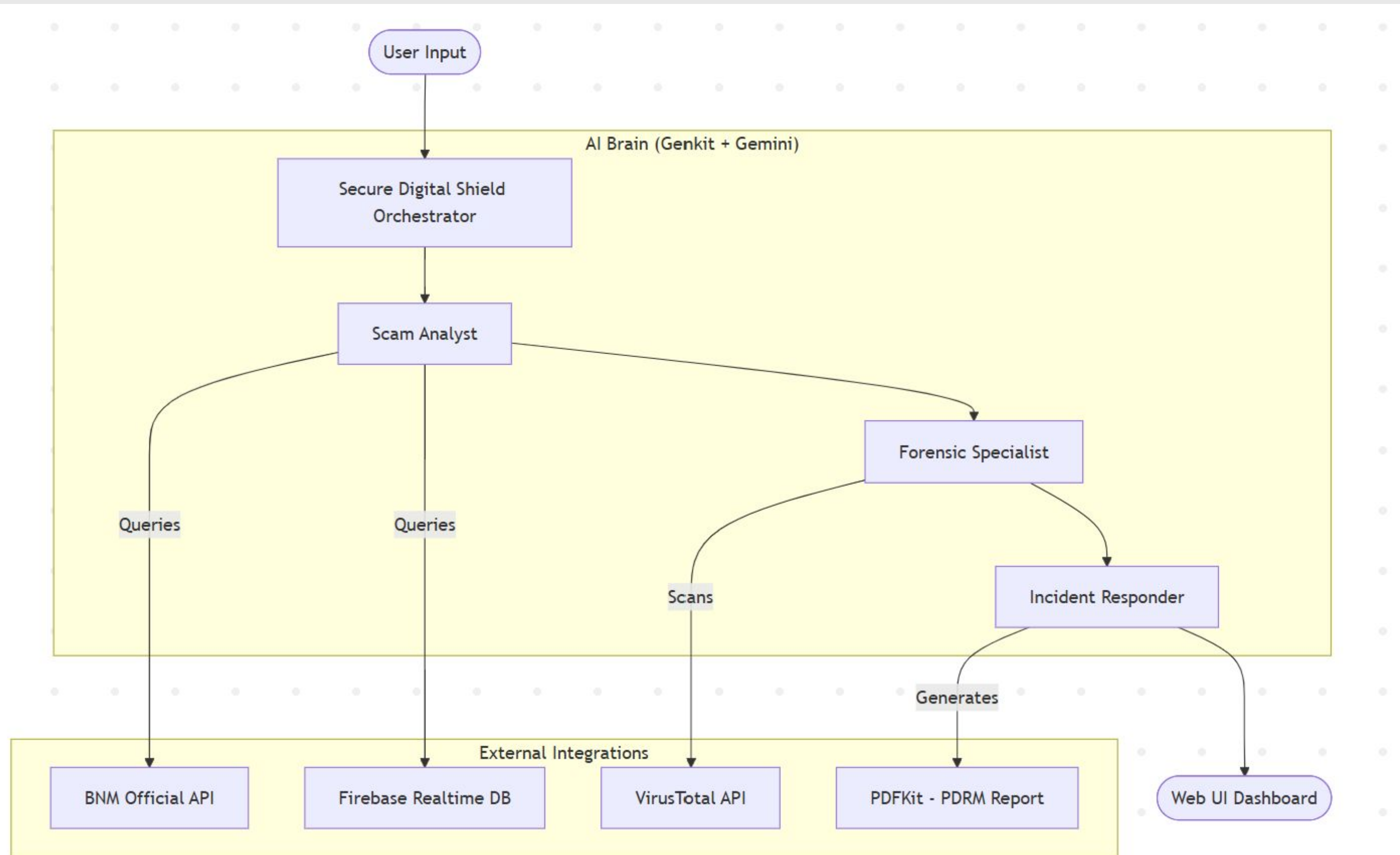
Using the Gmail API to "close the loop" by instantly notifying the bank or BNM during the critical first hour.



TECH STACK



AI AGENT WORKFLOW



CODE-AGENT 1:

Scam Analyst

```
// --- Helper: Fetch Live BNM Data ---
async function fetchBNMAlerts() {
  try {
    const url = "https://api.bnm.gov.my/public/consumer-alert";
    const response = await axios.get(url, {
      headers: { "Accept": "application/vnd.BNM.API.v1+json" }
    });
    return response.data.data.map((item) => ({
      name: item.name,
      websites: item.websites,
      source: 'BNM_OFFICIAL'
    }));
  }
  catch (error) {
    console.error("⚠ BNM API unreachable, proceeding with AI knowledge only.");
    return [];
  }
}
```

Finding Matching URL, Cases
Listed in BNM Database

```
// --- Helper: Fetch Internal Firebase Data ---
async function fetchFirebaseAlerts() {
  if (!rtdb)
    return [];
  try {
    const snapshot = await rtdb.ref('consumer-alert-datastore').limitToLast(100).once('value');
    const data = snapshot.val();
    if (!data)
      return [];
    return Object.values(data).map((item) => ({
      name: item.name,
      websites: item.websites || [],
      source: 'INTERNAL_FIREBASE_REPORTS'
    }));
  }
  catch (error) {
    console.error("⚠ Firebase RTDB fetch failed in ScamAnalyst:", error);
    return [];
  }
}
```

Finding Matching Cases
Listed in Firebase Data

CODE-AGENT 2:

Forensic Specialist

```
// Function to scan URLs using the URL Scanner tool
async function scanMaliciousUrls(urls, onProgress) {
  const scanResults = [];
  for (const url of urls) {
    try {
      if (onProgress)
        onProgress(`Scanning URL in VirusTotal: ${url}`);
      console.log(`🔗 Scanning URL with scanner tool: ${url}`);
      const result = await scanAndReportUrl(url);
      if (result) {
        scanResults.push({
          url,
          verdict: result.verdict,
          malicious: result.malicious,
          suspicious: result.suspicious,
          totalEngines: result.totalEngines,
        });
      }
    } catch (error) {
      console.error(`❌ Error scanning ${url}:`, error);
    }
  }
  return scanResults;
}
```

Scan to Check Viability of the URL Link through VirusTotal API

CODE-AGENT 3:

Incident Responder

```
//Export report in PDF Format
export async function buildPoliceReportPdf(report, outputFileName = 'PDRM_Police_Report.pdf') {
  const outputPath = path.resolve(process.cwd(), outputFileName);
  const doc = new PDFDocument({ size: 'A4', margin: 50 });
  const stream = fs.createWriteStream(outputPath);
  doc.pipe(stream);
  // Title
  doc.fontSize(20).font('Helvetica-Bold').text('PDRM Police Report', { align: 'center' });
  doc.moveDown(1);
  // Report Information Section
  doc.moveDown(0.5);
  doc.font('Helvetica').fontSize(12);
  doc.text('State: _____');
  doc.text('\n');
  doc.text('District: _____');
  doc.text('\n');
  doc.text('Contingent: _____');
  doc.text('\n');
```

Export the Police Report into a Downloadable PDF Format

SOCIAL IMPACT

DIGITAL TRUST

Enhances cybersecurity resilience which align with the Cyber Security Act 2025



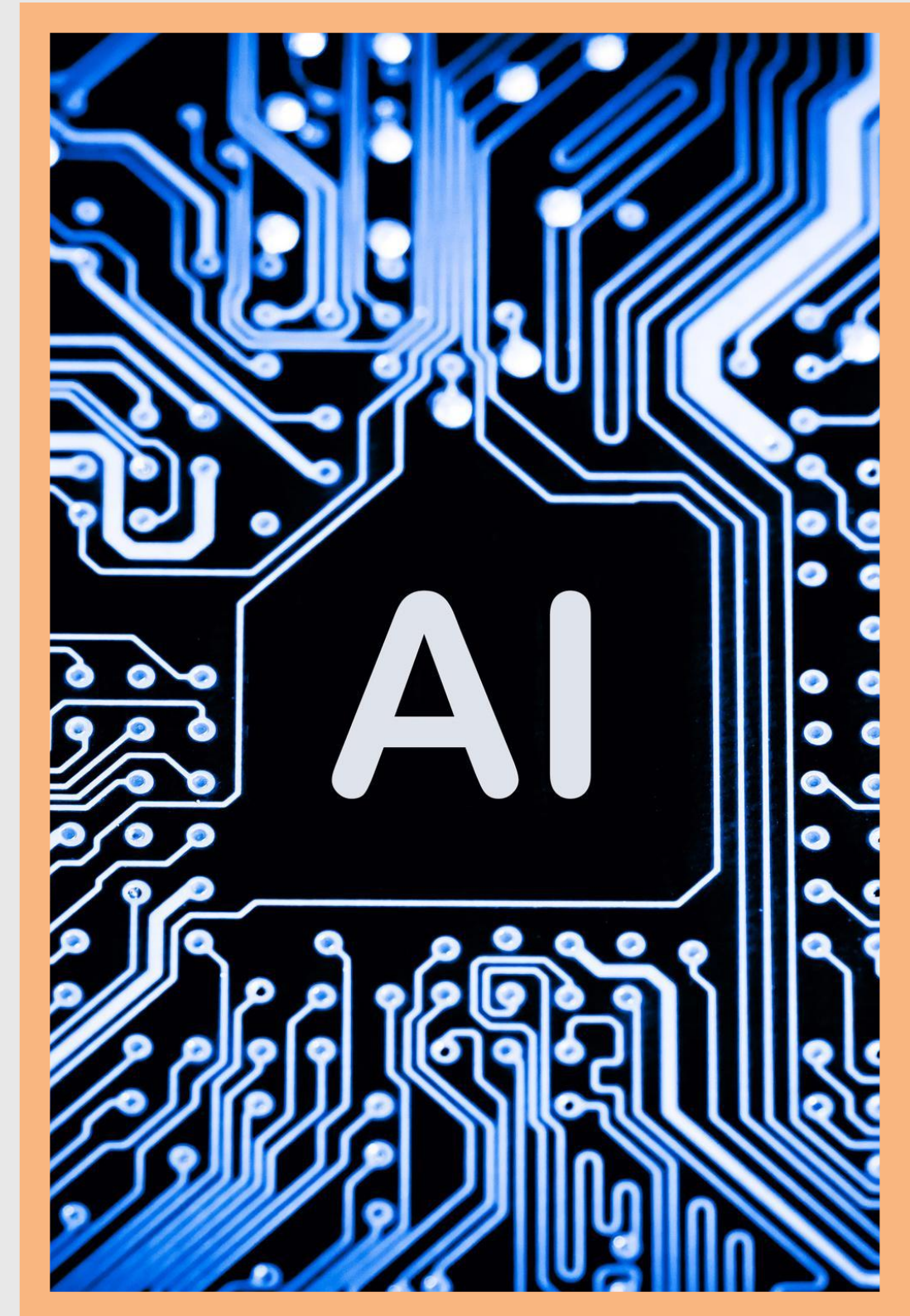
FINANCIAL WELL-BEING

Protects citizen assets from fraud, a goal of BNM Financial Sector Blueprint 2022–2026 .



AI GOVERNANCE

Implements ethical/safe AI which aligns with AI Governance Bill 2026

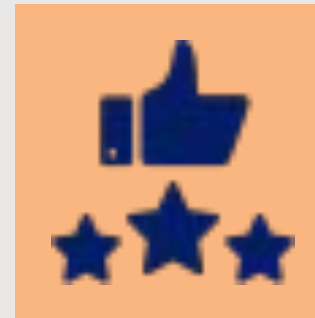


CHALLENGES



DATA ACCESSIBILITY

Due to strict privacy regulations, we operated without direct access to sensitive government datasets.



RIGOROUS DATA VALIDATION

A significant portion of our development was dedicated to sourcing and verifying high-fidelity, relevant datasets.



TECHNICAL ALIGNMENT

Problem in turning a complex requirement into a strategic advantage for our solution's scalability with Google technologies.

BUSINESS MODEL

Key Partners • PDRM & BNM: For "Ground Truth" data. • Google Cloud: Infrastructure partner. • MDEC/CyberSecurity Malaysia: For policy alignment.	Key Activities • AI Agent R&D: Refining Genkit & Vertex AI. • Data Orchestration: Syncing real-time scam blacklists.	Value Propositions • Forensic Accuracy: Verifying scams against official data. • Agentic Reasoning: Interactive AI analysis. • National Sovereignty: MADANI-aligned digital defense.	Customer Relationships • Trust & Transparency: Secure identity via login. • Automated Support: 24/7 AI-first responder. • Social Impact: Public safety focus.	Customer Segments • B2C: Everyday Malaysians (Elderly/B40). • B2B: Banks & E-wallets (TNG, CIMB). • B2B: E-commerce (Shopee/Lazada).
	Key Resources • Tech Stack: Vertex AI, Cloud Run, Firestore. • Data Sets: NSRC & VirusTotal APIs. • Human Capital: AI/Data Engineers.		Channel • Web/Mobile Platform: Direct user access. • S.O.F.A.R. API: Integrated into banking apps. • UTM/MDEC: Public awareness programs.	
Cost Structure • Cloud Infrastructure: Google Cloud usage fees. • API Licensing: Usage credits for forensic tools (VirusTotal, IP Intel). • On-Demand Compute: Scalable, pay-per-use hosting via Google Cloud Run.		Revenue Streams • SaaS API Licensing: B2B fees for banks/fintechs. • Freemium Plus: Detailed "Deep Forensic Reports." • Gov. Funding: MADANI social-innovation grants.		

THANK YOU

